

MATSUMOTO, Japan

WMO: 476180

Lat: 36.2467N

Lon: 137.9706E

Elev: 2007

StdP: 13.66

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.4	19.8	6.6	8.3	30.0	9.3	9.5	29.1	18.0	43.6	16.4	41.9	3.1	10	0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	17.8	92.0	72.8	89.6	72.0	87.0	71.1	74.7	87.0	73.7	85.3	72.7	83.3	8.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.5	125.2	79.2	70.4	120.4	78.6	69.5	116.7	78.0	39.8	87.5	38.8	86.0	37.8	83.3	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.7	15.0	13.4	DB	12.2	95.6	2.3	1.9	10.5	96.9	9.1	98.1	7.8	99.2	6.2	100.6
			WB	11.1	76.8	2.2	0.9	9.5	77.4	8.2	77.9	7.0	78.4	5.4	79.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.9	31.7	33.6	40.9	52.0	62.6	69.3	76.9	78.5	69.9	58.2	46.6	37.0
	DBStd	17.08	4.90	5.82	6.50	7.20	5.33	4.64	4.32	4.11	6.18	5.95	5.71	5.35
	HDD50	1927	567	460	294	61	1	0	0	0	0	6	134	404
	HDD65	4842	1032	879	748	391	109	14	0	25	224	552	868	
	CDD50	3714	0	1	11	122	390	580	834	885	597	261	32	1
	CDD65	1155	0	0	0	2	34	144	369	420	172	14	0	0
	CDH74	8743	0	0	0	47	370	907	2874	3494	994	57	0	0
	CDH80	3043	0	0	0	5	69	188	1083	1428	266	4	0	0
	WSAvg	5.3	4.9	5.1	6.0	6.3	6.1	5.2	5.4	5.2	4.7	4.5	4.9	5.2
	PrecAvg	40.3	1.3	1.6	2.8	3.2	3.8	5.5	5.3	3.7	5.8	3.9	2.0	1.1
(15) Precipitation	PrecMax	55.3	4.6	4.6	6.0	7.6	10.7	12.6	14.4	8.9	12.0	15.3	6.7	3.3
	PrecMin	25.2	0.0	0.2	0.2	0.4	1.3	1.4	0.9	0.3	0.7	0.2	0.2	0.0
	PrecStd	7.7	1.1	1.1	1.3	1.5	1.8	2.4	3.0	2.1	2.9	2.6	1.3	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	53.2	59.2	68.7	79.8	86.0	88.0	94.4	95.2	90.2	79.6	68.6	59.0	
	0.4% MCWB	43.9	46.8	49.0	56.0	60.8	66.6	73.3	73.5	70.8	64.6	54.6	49.1	
	2% DB	48.1	53.6	62.4	73.8	81.6	84.7	91.8	92.7	86.3	74.9	64.4	54.1	
	2% MCWB	39.5	43.4	46.7	53.8	60.1	66.3	73.0	73.0	70.3	62.2	52.8	45.3	
	5% DB	44.4	48.7	58.1	69.8	78.1	81.9	89.3	90.3	83.0	71.8	60.8	50.7	
	5% MCWB	36.9	39.8	45.3	52.6	58.3	65.4	72.1	72.6	69.0	60.9	51.4	42.9	
	10% DB	41.1	44.9	54.0	65.9	74.8	79.5	86.4	87.9	79.8	68.5	57.6	47.4	
	10% MCWB	34.4	37.1	43.6	51.4	57.6	64.9	71.4	72.2	68.0	58.9	49.8	40.4	
	0.4% WB	46.5	50.0	53.5	62.1	66.5	71.9	76.3	76.3	73.4	67.5	58.4	52.2	
	0.4% MCDB	51.1	56.3	59.7	67.6	75.8	81.7	89.5	89.3	84.4	75.6	63.7	55.9	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	40.6	44.7	50.0	57.7	64.1	70.1	74.8	75.0	71.8	64.9	55.6	47.1	
	2% MCDB	46.0	52.1	57.5	68.8	74.6	78.2	87.0	87.7	81.8	72.0	60.7	52.3	
	5% WB	37.3	40.8	47.2	55.2	61.8	68.6	73.7	74.1	70.7	62.7	53.2	43.6	
	5% MCDB	43.8	47.2	56.1	65.5	72.8	77.3	85.2	85.9	79.5	69.6	58.9	49.3	
	10% WB	34.7	37.5	44.3	52.9	59.9	67.1	72.6	73.2	69.5	60.5	50.6	40.7	
	10% MCDB	40.2	44.1	53.1	63.0	70.5	76.1	83.1	84.3	77.4	66.6	56.6	46.5	
(35) Mean Daily Temperature Range	MDBR	16.6	18.2	20.4	22.5	22.0	18.3	16.8	17.8	16.4	17.7	18.6	16.8	
	5% DB MCDBR	21.1	24.9	28.1	29.0	28.1	22.6	21.1	21.4	19.7	20.7	23.5	21.3	
	5% DB MCWBR	13.8	15.7	15.3	12.9	11.0	8.3	7.0	6.7	7.3	9.8	13.6	14.0	
	5% WB MCDBR	19.2	21.9	23.3	22.5	22.4	16.9	18.9	19.1	17.0	16.6	18.6	19.1	
	5% WB MCWBR	13.5	15.2	15.0	12.2	10.7	7.4	7.0	6.7	6.8	8.7	12.3	13.7	
(40) Clear-Sky Solar Irradiance	taub	0.366	0.409	0.481	0.526	0.544	0.580	0.561	0.565	0.490	0.435	0.385	0.361	
	taud	2.044	1.946	1.843	1.834	1.845	1.822	1.941	1.922	2.078	2.145	2.212	2.162	
	Ebn at Noon	251	254	247	244	242	232	236	231	243	245	246	246	
	Edn at Noon	40	50	61	65	66	67	59	59	48	41	34	34	
(44) All-Sky Solar Radiation	RadAvg	689	935	1216	1484	1667	1515	1485	1504	1211	1002	755	606	
	RadStd	50	60	84	125	175	107	216	149	101	96	60	58	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page